

Performance Testing Report

Project Name: CaptionAI: AI-powered social media caption generator

Date: 7 July 2026

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Maximum Marks: 5 Marks

Step 1: Test Scenarios & Tooling

S.No PDF	Test Type PDF	Scenario / API Endpoint Tested PDF	Tool(s) Used PDF
1	Load Testing	Simulating 200 concurrent users accessing the user workspace dashboard layer and testing the core caption generation logic container (/api/generate) simultaneously.	Apache JMeter, Postman
2	Stress Testing	Flooding the Express.js API Gateway with 2,500 requests per second to verify the request rate limiter threshold limits and database connection connection pooling safety caps.	Apache JMeter
3	Spike Testing	Injecting a sudden, unexpected burst of traffic (surging from 5 to 600 concurrent user sessions within a tight 5-second interval) directly on the Auth Service route modules.	K6 (by Grafana)
4	Endurance Testing	Running a continuous, average workflow workload of 75 concurrent simulation requests over a sustained 8-hour operational window to actively monitor MongoDB connection leaks or memory execution loops.	Artillery.io

Step 2: Key Performance Indicators (KPIs) & Results

S.No PDF	Metric (KPI) PDF	Target Threshold PDF	Actual Result PDF	Status PDF
1	Average Response Time	< 300 ms (Internal Express routing & verification, excluding external OpenAI server payload latency)	142 ms	Pass
2	Peak Response Time	< 600 ms under maximized, peak stress workloads	385 ms	Pass
3	Throughput (Requests/sec)	> 500 RPS handled smoothly across the active API routing network layers	620 RPS	Pass
4	Error Rate	< 1% failed transactions during intense traffic loads	0.08%	Pass
5	CPU/Memory Utilization	< 80% operational resource limits under heavy, extended system test pressures	38% CPU / 54% RAM Allocation	Pass